

Mihal Dimo

+39 3429491241 – Italy – mihal@kakao.com – github.com/mhdimo – mhdimo.github.io – LinkedIn

EDUCATION

University of Catania	September 2021 – Present (Expected June 2027)
Bsc in Computer Science	
Relevant Modules: Embedded systems, Algorithm & Data Structures, Linear Algebra	
Brandenburgische Technische Universität Cottbus	2024 – 2024
Bsc in Computer Science	Exchange Study Erasmus+
Relevant Modules: Applied Linear Algebra for AI, Calculus, Software Security	

WORK EXPERIENCE

Open Source Contributor	July 2026 – Present
vLLM Project	
- Contributed to 20+ merged PRs in the vLLM Metal backend, spanning speculative decoding, paged KV-cache management, model compatibility, sampling, and memory management for Apple Silicon inference. - Implemented DSpark speculative decoding for the vLLM Metal backend, including its Qwen3 model integration, scheduler-owned KV-cache management, paged execution, Markov drafting, prefix caching, validation, and serving benchmarks. - Optimized speculative decoding by eliminating redundant KV re-ingestion and chunking cold draft-KV ingestion, reducing 8K first-propose latency $31\times$ (3.47s $\rightarrow$ 112ms), E2E latency 2.1$\times$ (64.0 $\rightarrow$ 30.8 ms/token), steady-state KV ingestion by 50%, and DSpark draft-weight memory by 48% (4.42 $\rightarrow$ 2.29 GiB).	

Software Development Engineer Intern	January 2026 – July 2026
Amazon	Berlin, Germany
- Engineered a full-stack embedding search and visualization platform over 50M embeddings, with PCA/UMAP visualization and sub-5 ms KNN queries, accelerating Music recommender experimentation. - Designed a batch ML data pipeline for embedding metadata hydration and model indexing across S3, ECS, and OpenSearch, enabling automated weekly model updates. - Built evaluation workflows measuring marketplace performance with Recall, NDCG, Precision, and triplet accuracy, alongside a FastAPI backend and CloudWatch monitoring.	

Firmware Lead & Embedded Software Engineer	March 2024 – December 2025
KBDFans (Partner)	Changzhou, Jiangsu, China
- Lead firmware and hardware development for the Zellia Hall Effect project (team of 5 C/C++ engineers), designing a distributed embedded system across 5 AT32 MCUs with enhanced modularity and multi-layout keyboard using KiCad 8.0. - Achieved <0.28 ms input latency with 106 kHz Scan Rate by decoupling signal acquisition and processing, and engineered a custom 7.5 Mbps UART protocol for high-throughput multi-MCU synchronization. - Offloaded ADC normalization and calibration to slave MCUs, reducing master CPU load by 90% and enabling faster processing.	

PROJECTS

Custom FP8 CUDA Kernel – Qwen3.8-27B on H100	github.com/mhdimo/qwen38-h100-lab
- Wrote an SM90 CUDA kernel fusing SiLU, gating, and per-group FP8 E4M3 quantization, using warp-shuffle-only reductions with no block barriers; verified byte-exact against vLLM’s kernel across 168 reference comparisons (edge tiles, both scale layouts) and validated CUDA graph replay with changing inputs. - Profiled the serving stack to target the kernel (14.5% of prefill GPU time), reaching 2.26$\times$ kernel speedup at the 16K-row shape and +5.90% prefill / +3.65% mixed full-model throughput vs. tuned vLLM.	
Inference Engine — C++20 Metal LLM Runtime	github.com/mhdimo/inference-engine
- Built a LLM inference engine for Apple Silicon, implementing tensor runtime, compute graphs with lifetime-based memory planning, GGUF loading, tokenization, KV caching, sampling, and CPU/Metal backends. - Implemented 13 custom Metal compute kernels with zero-copy unified-memory execution, including tiled and quantized GEMM, RoPE, GQA attention, RMSNorm, and SwiGLU, enabling end-to-end inference of Qwen2.5 and SmolLM2.	

TECHNICAL SKILLS

Languages: C, C++, Rust, Python, TypeScript, Java, SQL.
ML Systems: vLLM, MLX (custom Metal kernels), PyTorch, Metal/MPS backends, speculative decoding, quantization, KV-cache optimization, embeddings, recommender systems, DQN.
LLM Tooling: MCP, multi-provider LLM APIs, AI-SDK, Claude code.
Systems & Data: Linux, Docker, AWS, REST APIs, Kafka, PySpark, OpenSearch, Parquet, Pandas, Numpy.
Embedded & Web: ARM Cortex-M, KiCad, USB/WebUSB, FastAPI, Node.js, SvelteKit, React.